



An Easy Access to Vinylboronates β -Substituted by a Keto Group.

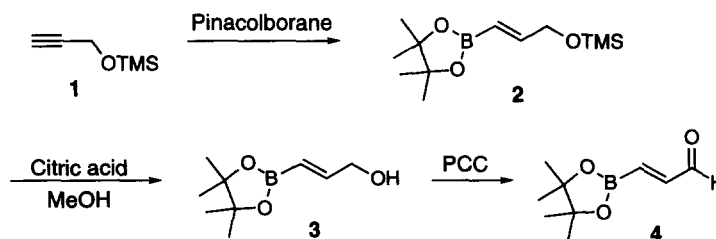
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Abstract : An easy access to 3-borylacrolein **4** is described. This α,β -unsaturated aldehyde was smoothly reacted with organozinc or organocuprates to give the corresponding allyl alcohols **5** in good yields. The α,β -unsaturated ketones **6** were easily obtained by oxidation of **5** with PCC. This overall sequence constitutes a general synthesis of these hitherto interesting new compounds.

E-Vinylboronates β -substituted with an electron withdrawing group are electron deficient olefins. These highly functionalized molecules offer an interesting synthetic potential as reactive dienophiles and radical acceptors.¹ Furthermore, the chemistry of the carbon-boron bond and the functional group transformations of the electron-withdrawing group offer supplementary synthetic possibilities. For example, 3-borylpropenoic acid derivatives are synthetic equivalents of E- β -hydroxyacrylic acid and E- β -hydroxyvinylamine.¹ Another member of this class of new electron poor olefins, i.e. 3-borylacrolein **4** was recently described.² That this olefin is a reactive dienophile was shown by its reaction with cyclopentadiene which occurred within 30 hours at -10°C giving a 9 : 1 mixture of endo and exo adducts.²

The preparation of **4** from 1,1-diethoxy-2-propyne led to purification problems which could be solved by the isolation and easy recrystallisation of the corresponding boronic acid. It was also shown that the addition of organolithium and magnesium derivatives led to the allylic alcohols **5** with modest yields as a result of a competitive nucleophilic addition on the boron atom of the boronic ester.² In this letter we describe a new efficient route to 3-borylacrolein **4**, the highly selective addition of organozinc and cuprates on the aldehyde function and a general access to α,β -unsaturated ketones β -substituted by a boronate ester. 3-Borylacrolein **4** was easily obtained on a quarter of a mole scale from propargyl alcohol according to scheme 1.

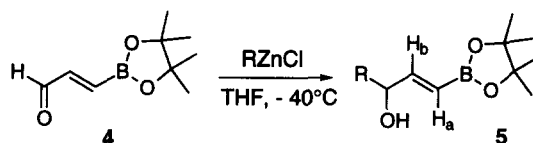


Scheme 1

This sequence begins with the hydroboration of the trimethylsilyl ether **1** easily prepared from propargyl alcohol with a slight excess of pinacolborane ³ which led to the vinylboronate **2** (bp_{0.2} mm/Hg = 85°C, 82 % yield). Deprotection of the alcohol function was effected with citric acid in methanol ⁴ giving the allylic alcohol **3** in 70 % isolated yield. A simple oxidation with PCC in dichloromethane at room temperature ⁵ gave the analytically pure aldehyde **4** in 81 % yield after bulb to bulb distillation and 46 % overall yield from propargyl alcohol.

3-Borylacrolein **4** presents three electrophilic sites which may be involved whenever a nucleophilic reagent is added to this molecule. We thought that organozinc derivatives would be the reagents of choice for a selective addition to the carbonyl group.⁶

Organozinc compounds obtained from the transmetalation of organolithium or organomagnesium derivatives with anhydrous ZnCl₂ at -80°C in THF reacted smoothly with 3-borylacrolein **4** at -40°C to give the allylic alcohols **5** in good yields after work up ⁷ (scheme 2, table 1).



Scheme 2

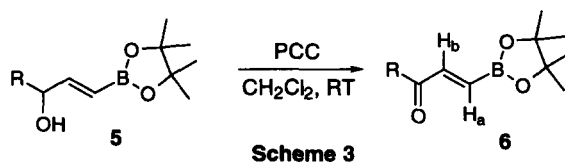
Table 1 : Addition of organozincs and mixed cuprates on the aldehyde **4**.⁸

R	Alcohol 5	JH _a H _b (Hz)	Yield % ^a
Me	5a	18.1	68
Et	5b	18.0	75
ⁿ Bu	5c	18.0	71
Ph	5d	17.9	60
allyl	5e	18.1	67
 CO ₂ Et	5f	18.0	73

^a Yields are for isolated pure products.

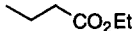
No product resulting from a nucleophilic addition on boron could be detected in the crude reaction mixture. Compounds **5e** and **5f** resulted from the addition of the mixed cuprates obtained by addition of one equivalent of CuCN, 2LiCl to the corresponding organozinc derivatives obtained from allyl bromide and ethyl β-bromopropionate.⁹ No 1,4-addition could be detected with these cuprates. All the allyl alcohols have the E configuration as shown by the JH_aH_b values (17.9 ≤ JH_aH_b ≤ 18.1 Hz).

We next turned to the selective oxidation of the allyl alcohols **5** which should give a general entry to the α , β -unsaturated ketones **6** according to scheme 3.



When treated with Dess-Martin's periodinate¹⁰ or under the Swern oxidation conditions,¹¹ the allyl alcohols **5** led to mediocre yields of ketones **6**. Although boranes are known to be oxidized by pyridinium chlorochromate (PCC),¹² the reaction of boronates **5** with this reagent at room temperature in dichloromethane led exclusively to the (E)- α , β -unsaturated ketones **6** in good yields as shown in table 2 ($J_{\text{H}_a\text{H}_b} \sim 18.5$ Hz).

Table 2 : Synthesis of the α , β -unsaturated ketones **6**¹³

R	Ketone 6	$J_{\text{H}_a\text{H}_b}$	Yield % ^a	bp°C/mm Hg ^b
Me	a	18.6	86	60-65/0.1
Et	b	18.5	89	110-112/0.2
nBu	c	18.5	90	c
Ph	d	18.4	75	90-95/0.2
allyl	e	18.5	81	120-125/0.2
 CO ₂ Et	f	18.4	72	100-105/0.2

^a : Yields of isolated pure products.

^b : Oven temperature is given (bulb to bulb distillation).

^c : Purified by flash chromatography over silica gel (eluant : CH₂Cl₂).

In conclusion, we have described a new and simple route to pure 3-borylacrolein **4**. Organozinc derivatives and cuprates add selectively on the carbonyl group leading to the corresponding allyl alcohols **5** in good yields. The selective oxidation of **5** led to vinylboronates **6**. This sequence represents a general access to this new family of electron poor olefins i.e. the vinylboronates β -substituted by a keto group. The chemistry of allyl alcohols **5** and enones **6** is under active investigation in these laboratories.

References and notes

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- 8 The procedure for the synthesis of the alcohol **5c** is representative. To a stirred solution of 272 mg of anhydrous ZnCl_2 (2 mmol) in anhydrous THF at -80°C was added 1.25 mL of a 1.6 M solution of *n*-butyllithium (2 mmol) in hexanes. The mixture was allowed to reach -40°C and then 273 mg (1.5 mmol) of 3-borylacrolein **4** in anhydrous THF were added. The solution was allowed to reach room temperature while stirring (3 h) and then the solvent was removed under vacuum. The crude product was partitioned between 4 mL of water and ether (20 mL). The aqueous phase was washed two times with 5 mL of ether. Organic phases were combined, dried over MgSO_4 , filtered and ether was removed under vacuum. The crude product was purified by bulb-to-bulb distillation (oven temperature : 110°C ; 0.2 mm Hg) : 256 mg of a yellow oil were obtained (71 % yield). ^1H NMR (CDCl_3 , 300 MHz) δ ppm/TMS : 0.85 - 0.95 (m, 3H) ; 1.27 (s, 12H) ; 1.30 - 1.38 (m, 2H) ; 1.45 - 1.60 (m, 2H) ; 1.78 (s, 1H, OH) ; 4.10 - 4.20 (m, 1H) ; 5.62 (dd, H_b , $J = 18.1$ and 5.3 Hz) ; 6.62 (d, H_a , $J = 18.1$ and 0.7 Hz). ^{13}C NMR (CDCl_3 , 75.5 MHz), δ ppm/TMS : 14.0 ; 22.7 ; 24.7 ; 27.5 ; 41.2 ; 73.8 ; 83.3 ; 117.0 ; 155.4.
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- 13 The procedure for the oxidation of **5a** is representative. To a stirred solution of 323 mg of PCC (1.5 mmol) in 2 mL of anhydrous dichloromethane was added under nitrogen a solution of **5a** (198 mg, 1 mmol) in 1 mL of CH_2Cl_2 at 0°C . The mixture was stirred at room temperature for 2 h and then filtered. The black residue was extracted three times with 5 mL of ether and the combined organic phases were filtered over celite. Removal of ether and CH_2Cl_2 under vacuum gave a crude oil which was purified by a bulb-to-bulb distillation (oven temperature : $60-65^\circ\text{C}/0.1$ torr). 168 mg of a colorless oil were obtained (86 % yield). ^1H NMR (CDCl_3 , 300 MHz) δ ppm/TMS : 1.28 (s, 12H) ; 2.30 (s, 3H) ; 6.54 (d, H_b , $\text{JH}_a\text{H}_b = 18.6$ Hz) ; 6.80 (d, H_a , $\text{JH}_a\text{H}_b = 18.6$ Hz). ^{13}C NMR (CDCl_3 , 75.5 MHz), δ ppm/TMS : 24.7 ; 26.7 ; 84.1 ; 132.0 ; 146.5 ; 190. IR liquid film : $\nu_{\text{C=O}} = 1720\text{ cm}^{-1}$; $\nu_{\text{C=C}} = 1620\text{ cm}^{-1}$.